Harsha Apoorv Merapureddy

Stony Brook, NY | (631) 406-1822 | harshaapoorva164@gmail.com | linkedin.com/in/harshaapoorv | github.com/Harshaapoorv | harshaapoorv.com

Education

Stony Brook University — Stony Brook, NY
Master of Science in Computer Science (AI/ML focus) — *Aug 2026 – May 2028 (Expected)*
- **Coursework:** Artificial Intelligence, Data Science Fundamentals

Sagi Rama Krishnam Raju Engineering College (JNTU Kakinada) — Bhimavaram, India
Bachelor of Technology in Computer Science and Engineering; CGPA: 8.13/10 — *Apr 2023*

Experience

Stony Brook University — Stony Brook, NY
Graduate Teaching Assistant, CSE 312: Social, Legal, and Ethical Issues in Computing — *Aug 2026 – Present*
- Grade written assignments and exams for 2 sections (**120 students**) against course rubrics

Tata Consultancy Services (TCS) — Bengaluru, India
Systems Engineer / Mobile Application Developer — *Jun 2023 – Aug 2026*
- Built UK NEST Pensions app features in **React Native** and **JavaScript** from inception, with a **TypeScript** Redux store; 7 major and 3 minor releases (**100K+ downloads**, **5K+ DAU**)
- Cut load time **20%** via **lazy loading**; aided a mobile-first migration tied to a reported **40%** satisfaction gain
- Proposed and built **accessibility** features (text zoom, voice-out) to help older savers use the app
- Created reusable React Native components that sped up feature delivery by **nearly 50%**
- Implemented **FCM** push, **deep linking**, **AsyncStorage** offline storage, and **Keychain**-secured tokens
- Wrote **Jest** unit tests and strengthened team testing practices, contributing to **35% fewer** production defects
- Debugged Android native modules (Kotlin/WebView) during the React Native New Architecture migration
- Supervised **2 associates** for nearly 2 years; also interviewed candidates and presented to UK clients

Blackbucks — Remote
Web Development Intern — *Feb 2022 – Mar 2022*
- Built a resume builder web application using **HTML**, **CSS**, and **JavaScript**

Projects

NotesApp | *React Native, Redux Toolkit, RTK Query, Python, FastAPI, PostgreSQL* | GitHub: App, Backend
- Built and deployed solo: React Native client, layered **FastAPI** REST API (SQLAlchemy, Alembic), **PostgreSQL**
- Implemented **Google OAuth**/OTP sign-in and an RTK Query layer that rotates **JWT** refresh tokens on 401s
- Added a rich-text editor, Excel export, and a config API for force updates and maintenance mode

PassionIsle (Event Discovery App) | *React Native, FastAPI, PostgreSQL, Redis, AWS S3* — In Development
- Sole engineer: wrote the architecture doc, REST API spec, and requirements; designed all screens in Figma
- Designed **FastAPI**/**PostgreSQL** backend; **Redis** for OTP and rate limiting, **S3** media; now implementing
- Logged design trade-offs as architecture decision records (ADRs) and managed scope via formal change requests

Cyberbullying Detection and Reporting | *Python, Machine Learning, NLP, TF-IDF* — Undergraduate Capstone
- Led a 5-member team; built **TF-IDF** features and compared classifiers, reaching up to **92% accuracy**

Technical Skills

Languages: Python, JavaScript, TypeScript, SQL, HTML, CSS
Frontend & Mobile: React Native, React, Redux Toolkit, RTK Query, React Navigation, FCM, AsyncStorage
Backend & Data: FastAPI, REST APIs, SQLAlchemy, Alembic, Pydantic, PostgreSQL, Redis, Firebase
Tools & Concepts: Git, Jest, JWT, OAuth 2.0, AWS S3, Render, Data Structures & Algorithms, Machine Learning

Leadership & Activities

- **Secretary, Competitive Programming Club, SBU** (2026–27): contest coordination incl. ICPC regional
- **ACE, SRKR** (2021–23): Executive → Senior Body → Helping Hand; ran hackathons and fests (1,000+ attendees)
- **Athletics:** Captain, department football and basketball teams (SRKR); member, SBU competitive cricket team
- **Python Instructor (Online):** taught Python to 60+ students
- **TCS Recognition: 9 awards** (incl. Technical Excellence) and **7 appreciations** in 3 years
- **Awards:** HackerRank 5-Star (Python); CodinGame Legend (Coding Speed); #448, Newton's Coding Challenge 2020